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ORIGINAL ARTICLE

Detection of focal impaired awareness seizures using a biometric shirt

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Abstract

Objective: In recent years, seizure detection using wearable technology has gained significant attention in research. Most studies, however, have focused on detecting generalized or focal to bilateral tonic–clonic seizures. This study evaluates the feasibility of using a biometric shirt to detect focal impaired awareness seizures (FIAS) by monitoring respiratory, electrocardiography, and accelerometry signals.

Methods: Patients with epilepsy were recruited at the University of Montreal Hospital Center epilepsy monitoring unit. Seizures were annotated by epileptologists based on simultaneous video-electroencephalographic recordings, blinded to the shirt data. Features were extracted from the respiratory, accelerometry, and electrocardiography signals using varying window sizes and steps. An XGBoost classifier was trained and tested using a nested leave-one-subject-out cross-validation. Post-processing included a firing power regularization method to reduce false alarms.

Results: We recorded 113 FIAS from 27 patients who wore the shirt continuously for over 4750 hours. Using a firing power threshold of 0.65, we detected 71 seizures, resulting in a mean sensitivity of 66%, a 15% time in warning, and a false alarm rate (FAR) of 30 per 24 hours. A firing power threshold of 0.85 allowed us to reduce false alarms (8% time in warning, FAR of 21 per 24 hours) but resulted in a lower sensitivity of 57%. Performances varied across patients: sensitivity was high and FAR was low for some patients and vice versa for others, indicating variability in algorithm effectiveness across patients.

Significance: Our results demonstrate that detecting FIAS with a connected shirt could be feasible for certain patients, although the rate of false alarms remains an issue. Designing a personalized algorithm and selecting patients who exhibit significant physiological changes during seizures could make wearable-based FIAS detection more viable in the near future.

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Plain Language Summary: Novel mobile health technologies could transform epilepsy care by enabling continuous monitoring of seizures in everyday life. In this study, we used a biometric shirt to detect seizures with impaired awareness automatically. We used the shirt to measure breathing, heart activity, and movement in 27 patients with epilepsy in a hospital setting. Our algorithm detected up to two-thirds of seizures correctly. However, the number of incorrect alarms remains relatively high, with variable performances between patients. While the technology showed potential, these challenges highlight the need for further improvements and personalized care plans.

KEYWORDS

epilepsy, focal impaired awareness seizures, machine learning, physiological signals, seizure detection, time-series processing, wearables

1 | INTRODUCTION

Epilepsy is a chronic neurological condition characterized by recurrent spontaneous seizures due to abnormal and excessive neuronal discharges.[1,2] It affects over 50 million people worldwide.[3] While the first line of treatment consists of long-term drug therapy, 30%–40% of patients have seizures that are resistant to anti-seizure medications.[4] Refractory epilepsy poses significant challenges, including risk of injury, anxiety, and difficulty maintaining employment.[5]

There is a growing interest in the development of non-invasive seizure detection devices.[6] A reliable alert system could notify caregivers or emergency services and trigger timely interventions, potentially reducing risks associated with seizures.[7] Moreover, accurate seizure monitoring is crucial for evaluating the effectiveness of treatments, especially during periods of medication adjustment.[8] Indeed, seizure count currently relies on self-reported seizure diaries, which could be unreliable.[9]

While electroencephalography (EEG)-based seizure detection algorithms have shown promising performance, EEG is impractical for everyday use in an outpatient setting. Recent research investigated seizure detection using non-intrusive connected biometric devices that monitor physiological signals such as accelerometry,[10] photoplethysmography (PPG),[11] electrocardiography (ECG),[12] and electromyography (EMG).[13] These devices should be comfortable, non-stigmatizing, and user-friendly to ensure acceptance and regular use.[14]

Presently, commercially available connected devices focus primarily on detecting generalized tonic–clonic seizures (GTCS) and focal to bilateral tonic–clonic seizures (FBTCS).[15] For instance, the Embrace smartwatch detects convulsive seizures lasting more than 20 seconds,[16] and the EMFIT movement monitor is used for the detection of nocturnal convulsive seizures.[17] Only a few recent studies explored detecting other seizure types, such as focal impaired awareness seizures (FIAS).[15] For example, Böttcher et al.[18] developed a machine learning algorithm for detecting motor FIAS using a wrist-worn wearable device that records electrodermal activity (EDA), accelerometry, and blood volume pulse (BVP). Similarly, Jeppesen et al.[12] implemented a patient-specific FIAS detection algorithm based on heart rate variability. The study included data recorded with the ePatch heart monitor. The authors selected patients who experienced an increase of more than 50 beats/min in heart rate during seizures. Building on this, our research explores the feasibility of detecting FIAS using a biometric shirt that records raw ECG, respiration, and accelerometry. By including different types of FIAS, we aim to explore the broad applicability of our findings.

Key points

- A smart shirt was used to detect focal impaired awareness seizures (FIAS).
- Electrocardiography, accelerometry, and respiratory inductance plethysmography were collected from 27 patients in a clinical setting.
- Extracted features were used to train a machine learning algorithm for seizure detection, using a nested leave-one-out cross-validation.
- A sensitivity of 66% with a time in warning of 15% was obtained.
- For 48% of patients, all FIAS were correctly detected.

This work represents the first investigation into using a biometric shirt for multimodal FIAS detection.

2 | MATERIALS AND METHODS

2.1 | Patient recruitment

Patients admitted to the University of Montreal Hospital Center (CHUM) epilepsy monitoring unit (EMU) between April 2019 and May 2023 were prospectively recruited and asked to continuously wear the Hexoskin biometric shirt. The study received approval from the CHUM research ethics committee (18.091), and all participants provided informed consent prior to enrollment. Patients were simultaneously monitored using video–EEG, the gold standard for recording and annotating epileptic seizures.

Eligible participants were adult patients admitted to the EMU for at least two days. Exclusion criteria were patients with intracranial EEG, patients presenting a difficulty wearing the shirt due to a medical reason (e.g., orthopedic cast on upper body), and patients without an epilepsy diagnosis or with psychogenic non-epileptic seizures (PNES). Patients who did not experience FIAS during the study were excluded from the current analysis.

2.2 | Data acquisition

The Hexoskin biometric shirt was fitted to each patient according to the Hexoskin sizing chart. The shirt allowed recording of 3-axis acceleration (64 Hz), thoracic and abdominal respiratory inductance plethysmography (128 Hz), and 1-lead ECG (256 Hz) (Figure 1). Patients wore the shirt continuously, and daily visits were performed to change the recording device and apply moisturizing cream to improve ECG electrode contact. Data acquired with the Hexoskin shirt were previously validated by comparison against standard laboratory measurement devices.[19,20]

2.3 | Seizure annotation

Seizure onset and offset were annotated by three board-certified epileptologists (DHT, TPYT, and DKN) through a retrospective review of video-EEG monitoring, blinded to the shirt data. Seizure classification was based on the 2017 International League Against Epilepsy (ILAE) classification.[21] The video-EEG system and the Hexoskin device were both synchronized to the EMU server's clock, ensuring that the shirt signals' timestamps accurately corresponded with the seizure onset times annotated on video-EEG.

FIGURE 1 Hexoskin biometric shirt equipped with electrocardiogram sensors, thoracic and abdominal respiratory bands, a 3-axis accelerometer, and a recording device. Front (left) and inside out (right) views.

2.4 | Signal preprocessing and feature extraction

Features were extracted from continuous recordings using five different window sizes (10, 15, 20, 25, and 30 s) and three different steps (1, 3, and 5 s). This process transforms signals with different sampling frequencies into features, resulting in the same number of epochs for each signal (Figure 2).

2.4.1 | Respiration

Raw respiratory signals were filtered using the Daubechies wavelet ("db6") to extract the primary signal features and minimize noise, keeping only the first 3 components.[22] In addition, signals were bandpass filtered [0.05–3 Hz] using a second order Butterworth filter.[23] Breathing rate and amplitude were then extracted from both abdominal and thoracic respiratory bands (Figure 2).

2.4.2 | ECG

ECG signals were processed using a 0.5 Hz high-pass Butterworth filter (order 5). A 60 Hz notch filter was implemented using a moving average kernel (width of one period of 60 Hz). R-peaks were identified as local maxima within the absolute gradient of the filtered ECG signal.[24] A total of 19 non-linear and linear heart rate variability (HRV) features previously used for FBTCS detection[25] were extracted based on detected R-R intervals and are detailed in Table S1.

2.4.3 | Acceleration

A total of 62 acceleration features, which have shown promising performances for FBTCS detection, were extracted.[25] Fourteen features, described in Table S2, were computed for each of the X, Y, and Z axes, as well as for the norm of these three axes. Additionally, correlation coefficients between each pair of the X, Y, and Z acceleration axes were assessed. A complete list of all features is included in Table S2.

2.5 | Training and testing strategy

Due to the high imbalance between interictal and ictal data, interictal data was undersampled to include one hour

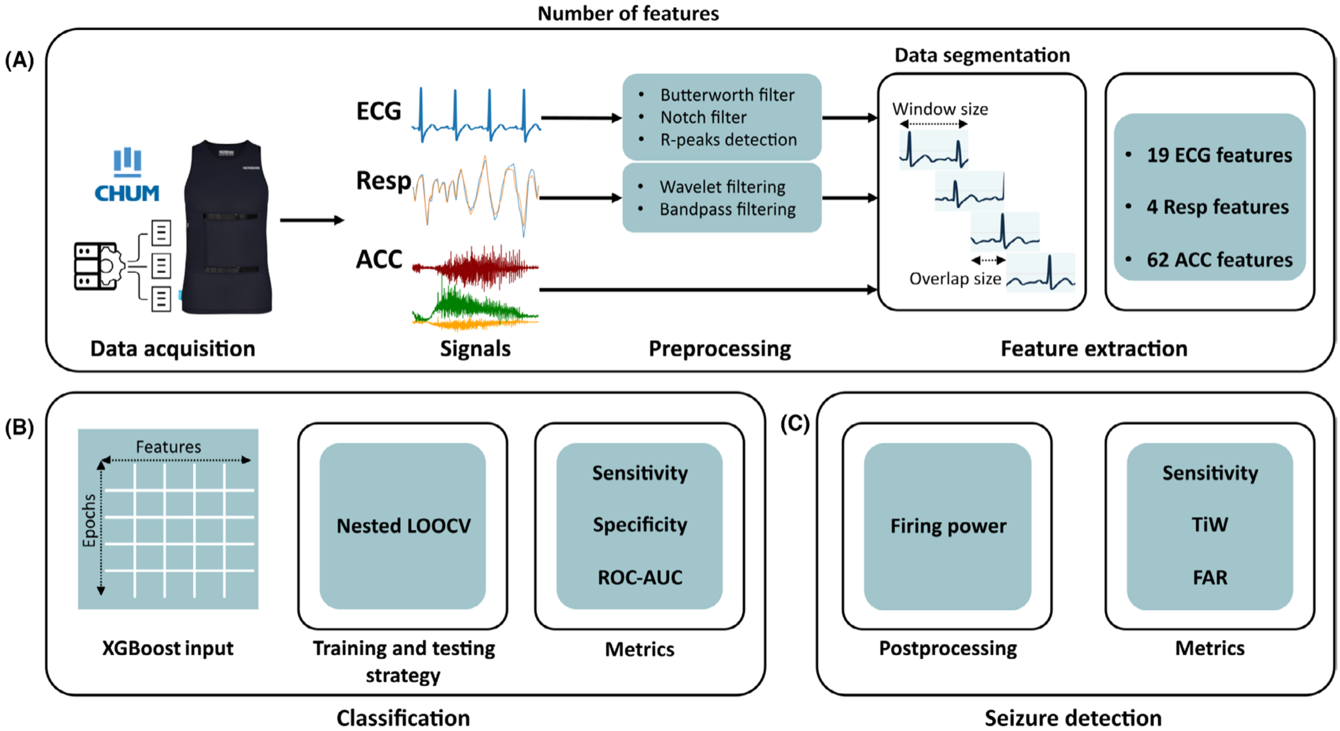

FIGURE 2 Data analysis pipeline with (A) Data acquisition and processing, using different window sizes (10, 15, 20, 25, and 30 s) and three step sizes (1, 3, and 5 s). Four respiratory, 19 ECG and 62 ACC features were extracted. (B) Machine learning classification using an XGBoost and (C) seizure detection post-processing and evaluation. ACC, Acceleration; ECG, Electrocardiography; FAR, False alarm rate; LOOCV, Leave-one-subject-out cross-validation; Resp, Respiration; ROC-AUC, Area under the receiver operating characteristic curve; TiW, Time in Warning.

of interictal data for each seizure event. This approach accelerates training while mitigating the risk of the classifier becoming biased towards the predominant interictal class. An eXtreme Gradient Boosting (XGBoost) classifier was trained for the binary classification of interictal and ictal epochs.[26] A nested leave-one-subject-out cross-validation (LOOCV) was implemented to evaluate performance for each patient (Figure 3). The outer loop (leave-one-subject-out) evaluates the model's performance by excluding one patient at a time. The inner loop performs five-fold cross-validation on the remaining patients to determine the optimal hyperparameters. This approach ensures a robust assessment of model performance and prevents overfitting by testing each hyperparameter combination on multiple data splits.

All performance metrics were reported on the test set using all data available for each held-out patient (at each iteration of the LOOCV outer loop), amounting to a total of over 4750 hours of recordings for the 27 patients.

In addition to window size and step, other XGBoost hyperparameters were optimized, including maximum depth[7,9,11] and minimum child weight.[5,7] These hyperparameters were chosen because they control the complexity of the model and prevent overfitting. Maximum depth limits how deep the tree can grow, which controls model complexity, while minimum child weight ensures that the model does not create overly specific nodes, providing a regularization effect.[26] Given the highly unbalanced data, the weight parameter of the positive class was adjusted to match the observed ratio in the patient group, ensuring that the model correctly emphasizes the minority class (seizure events).

2.6 | Post-processing

The XGBoost classifier generates binary predictions for each epoch. Detecting FIAS with wearable devices is challenging, often leading to many non-seizure epochs being misclassified as seizures. To reduce the false alarm rate (FAR), we implemented the firing power post-processing regularization method.[25] This approach triggers an alarm when the average of binary predictions within a 30-second window exceeds a predefined threshold. The alarm remains active until the average falls below this threshold. Furthermore, if another alarm occurs within 4 minutes of the previous one, the two alarms are merged. Increasing the threshold reduces the number of false alarms but at the expense of potentially missing seizures (Figure 2).

2.7 | Performance evaluation metrics

Given that our dataset is highly imbalanced, we avoided using metrics that cross-reference classes. Instead, for epoch-based classification, we evaluated performance

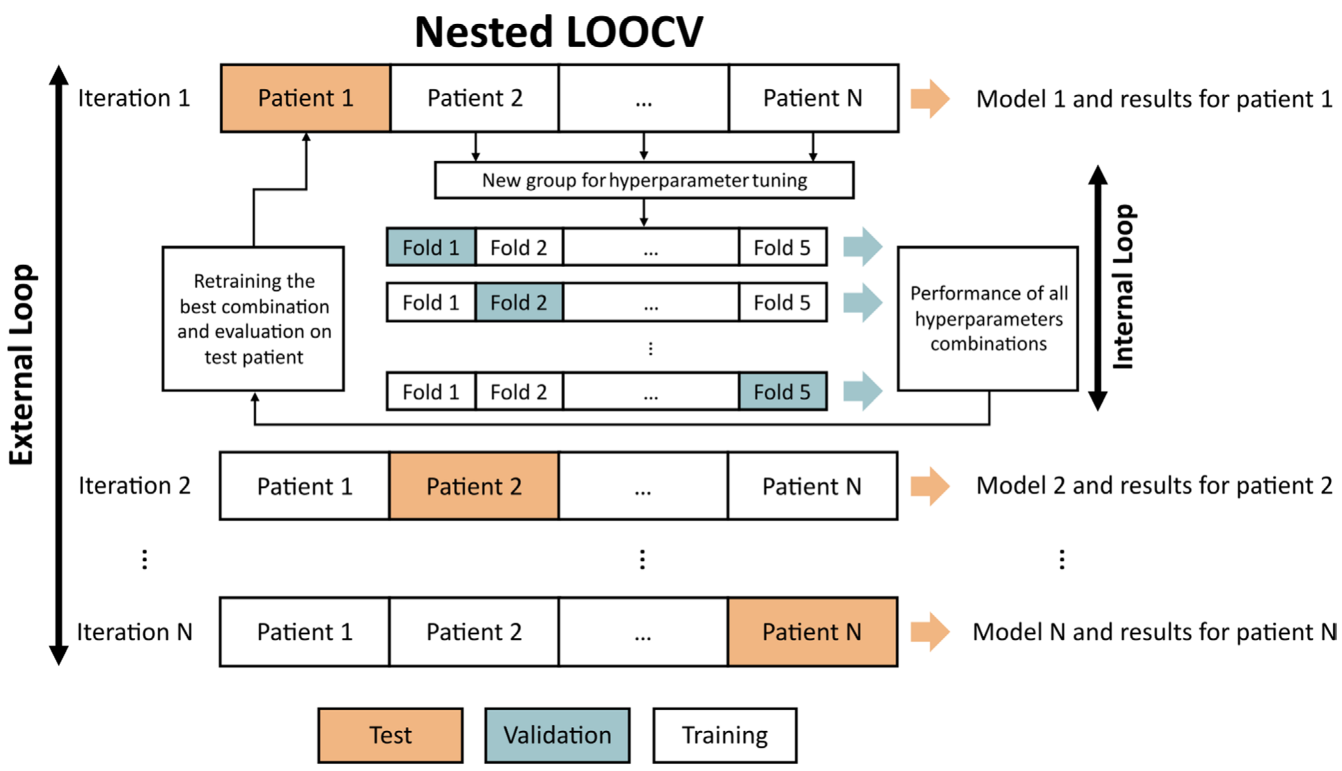

FIGURE 3 Nested leave-one-subject-out cross-validation (LOOCV) methodology.

in terms of sensitivity, specificity, and the area under the receiver operating characteristic curve (ROC-AUC) (Figure 2).

For event-based classification, performances were evaluated in terms of sensitivity, FAR per 24 hours (FAR/24 h), and the percentage of Time in Warning (TiW). Here, the sensitivity represents the percentage of detected seizures to the total number of seizures per patient, rather than the percentage of correctly classified seizure epochs.

3 | RESULTS

3.1 | Patient cohort

The dataset included 27 patients who continuously wore the Hexoskin shirt for more than 4750 hours and experienced a total of 113 FIAS. Table 1 presents patient demographics, total recording time per patient, number of recorded FIAS per patient, and other seizure types. In addition to FIAS, we annotated focal aware seizures (FAS), focal to bilateral tonic–clonic seizures (FBTCS), and focal unknown awareness seizures (FUAS). Patients' ages ranged from 19 to 61 years, with various, sometimes multifocal, epileptic foci. Ten patients were female. No adverse events related to the use of the shirt were reported by participants during the study period.

3.2 | Classification of interictal and ictal epochs

We trained models using different window sizes (from 10 to 30 s) with a varying step (1, 3 and 5 s). While the differences in performances across these configurations were minimal, the XGBoost classifier reached the highest performance in detecting FIAS when using a 10-second window with a 5-second step, as illustrated in Figure S1. While this optimal configuration attained the highest epoch-based sensitivity of 55%, an epoch-based specificity of approximately 90% was consistently achieved across all configurations. Average ROC-AUC for this model was 0.76. However, as illustrated in Figure S2, the performance varied significantly among individuals, with some achieving ROC-AUC scores as high as 0.95, while for others, the performance was notably lower.

As illustrated in Figure 4, epoch-based sensitivity varied greatly between patients, ranging from 0% to 100%, whereas epoch-based specificity consistently ranged from 70% to 95%. This variability is expected due to the variability in seizure semiology and the fact that seizures occupy a relatively small proportion of the total recording time compared to the interictal state.

3.3 | Seizure detection after post-processing

Using a 30-second window (6 epochs) for the firing power post-processing step, we tested thresholds of 0.65 and 0.85, respectively, corresponding to 4 and 5 positive classifications out of 6 consecutive epochs.

For the threshold of 0.85, 61 of 113 FIAS were detected, reaching an average sensitivity of 57% (Figure 5A), averaging 21 false alarms per 24 hours (Figure 5B; time in warning: 8.31%). Lowering the detection threshold to 0.65 increased the average sensitivity per patient to 66% with 71 detected FIAS (Figure 5A), but also increased the TiW to 14.91% (Figure 5B) and the average FAR to 30.6 false alarms per 24 hours. TiW was highly variable across patients. For 48% of patients (13/27 patients), an event-based sensitivity of 100% was achieved.

There were 5 patients for whom no FIAS were detected. Patient 7 had two 30-second seizures, with one seizure having no clinical manifestation and one seizure with upper limb tonic posture and then left hand automatisms. Patient 8 had 2 seizures, with a tonic posture and then myoclonic upper limb movements. Patient 13 had 10 seizures showcasing only behavior arrest for all of them. Patient 18 had two seizures with manual and oral automatisms. Patient 22 had only one seizure with oral-facial and manual automatisms.

Hence, the 5 patients with no detected seizures have seizure semiologies that are difficult to detect with ECG, respiration, and accelerometers. If we were to exclude those patients from the analysis, as they would be considered non-responders to the algorithm, the average sensitivity achieved for the 22 remaining patients would increase to 81%.

Four patients who experienced more than 2 seizures each had their seizures detected with 100% sensitivity. Patient 1 had 9 seizures with manual and oral automatisms. Patient 4 had 11 seizures, all with motor components, either upper limb automatisms, tonic posture, or hyperkinetic movements and associated tachycardia. Patient 9 had 4 seizures, all with behavioral arrest. Lastly, patient 11 had 6 seizures with oral and manual automatisms.

We also investigated potential differences in sensitivity between male and female patients. At a firing power threshold of 0.65, mean sensitivity was 62.6% for male patients ($n=17$) and 72.5% for female patients ($n=10$), compared to an overall mean sensitivity per patient of 66.3%. At a more conservative threshold of 0.85, sensitivity decreased to 58% for males and 55.4% for females. We found that using both firing power thresholds, sensitivity per sex groups was not normally

TABLE 1 Demographics of recruited patients.

Patient ID	Sex	Age at recruitment	Epileptic foci[a]	Total device recording time (h:min:s)	Number of recorded FIAS	FIAS semiology[b]	Other seizure types (number)
1	M	55	Bitemporal	196:30:35	9	Manual and oral automatisms	FAS (1)
2	M	28	Right frontal	97:11:29	4	Dystonic posture, UL automatisms, eye-blinking, and tachycardia	FAS (45)
3	M	22	Right insular	187:41:44	4	Oral and manual automatisms, UL clonia, and LL tonic posture	FAS (2), FBTCS (1)
4	M	61	Right temporal	208:00:32	11	Manual automatisms, tonic posture, hyperkinetic movements, and tachycardia	FAS (2)
5	M	33	Bitemporal	238:55:06	11	Tachycardia, bradycardia, behavior arrest, aphasia, oral, and manual automatisms	-
6	M	30	Left superior parietal	119:34:20	1	Dystonic posture, and pedaling	-
7	M	19	Left temporal	190:11:24	2	UL tonic posture, and manual automatisms	-
8	M	21	Left frontal	68:40:59	2	Tonic posture, UL myoclonia	FBTCS (1)
9	F	21	Left temporo-occipital	91:45:01	4	Behavior arrest	FAS (5), FBTCS (1)
10	M	20	Right frontal	118:10:33	2	Asymmetric tonic posture, and LL clonia	FBTCS (1)
11	M	23	Right temporal	183:01:33	6	Oral and manual automatisms	FBTCS (1)
12	F	25	Left fronto-temporal	254:28:33	7	Tonic posture, aphasia, oral automatisms, eye- blinking, laughing, and vocalization	FAS (3), FBTCS (3)
13	F	44	Left fronto-temporo-insular	19:09:53	10	Behavior arrest	–
14	F	40	Right fronto-temporal	138:16:13	2	Hyperventilation, behavior arrest, and dystonic posture	FAS (10)
15	M	45	Multifocal	197:22:29	2	Behavior arrest, manual automatisms, and UL dystonic posture	FAS (2)
16	F	19	Left temporal	199:56:24	1	Dysphasia	FUAS (12)

(Continues)

TABLE 1 (Continued)

Patient ID	Sex	Age at recruitment	Epileptic foci[a]	Total device recording time (h:min:s)	Number of recorded FIAS	FIAS semiology[b]	Other seizure types (number)
17	M	46	Right fronto-temporal	167:22:27	1	Sense of heat and hypersalivation	FAS (18)
18	M	52	Right temporal	190:05:03	2	Manual and oral automatisms	–
19	M	54	Right temporal plus	181:09:32	3	Hyperkinetic activity and oral and manual automatisms	–
20	F	19	Left temporal	234:23:25	1	UL automatisms	FBTCS (3)
21	F	50	Bitemporal	166:02:12	2	Oral and manual automatisms	–
22	M	28	Left temporal plus, right temporal	99:31:27	1	Oral-facial and manual automatisms	FAS (1), FUAS (1), FBTCS (3)
23	M	51	Bitemporal	402:23:24	1	Oral and manual automatisms and aphasia	FUAS (2)
24	F	45	Bitemporal	273:34:46	1	Tachycardia and oral-facial automatisms	–
25	F	31	Multifocal	250:01:34	2	Eye-blinking	FAS (6)
26	F	42	Multifocal	132:04:56	15	UL dystonia, and LL automatisms	–
27	M	27	Left temporal plus	175:06:58	6	Behavior arrest and UL automatisms	FBTCS (3)
Total	M (17), F (10)	–	–	4780.5	113	–	FBTCS (17), FUAS (14), FAS (95)
Median (range)	–	31 (19–61)	–	183 (19–402)	2 (1–15)	–	–

Abbreviations: F, female; FAS, focal aware seizure; FBTCS, focal to bilateral tonic–clonic seizure; FIAS, focal impaired awareness seizure; FUAS, focal unknown awareness seizure; h, hour; LL, lower limb(s); M, male; min, minute; s, seconds; UL, upper limb(s).

[a]Epileptic foci were not validated by intracranial electroencephalographic recordings.

[b]Reported semiology represents the predominant clinical descriptors observed during focal impaired awareness seizures in each patient. Semiology may vary between individual seizures for the same patient.

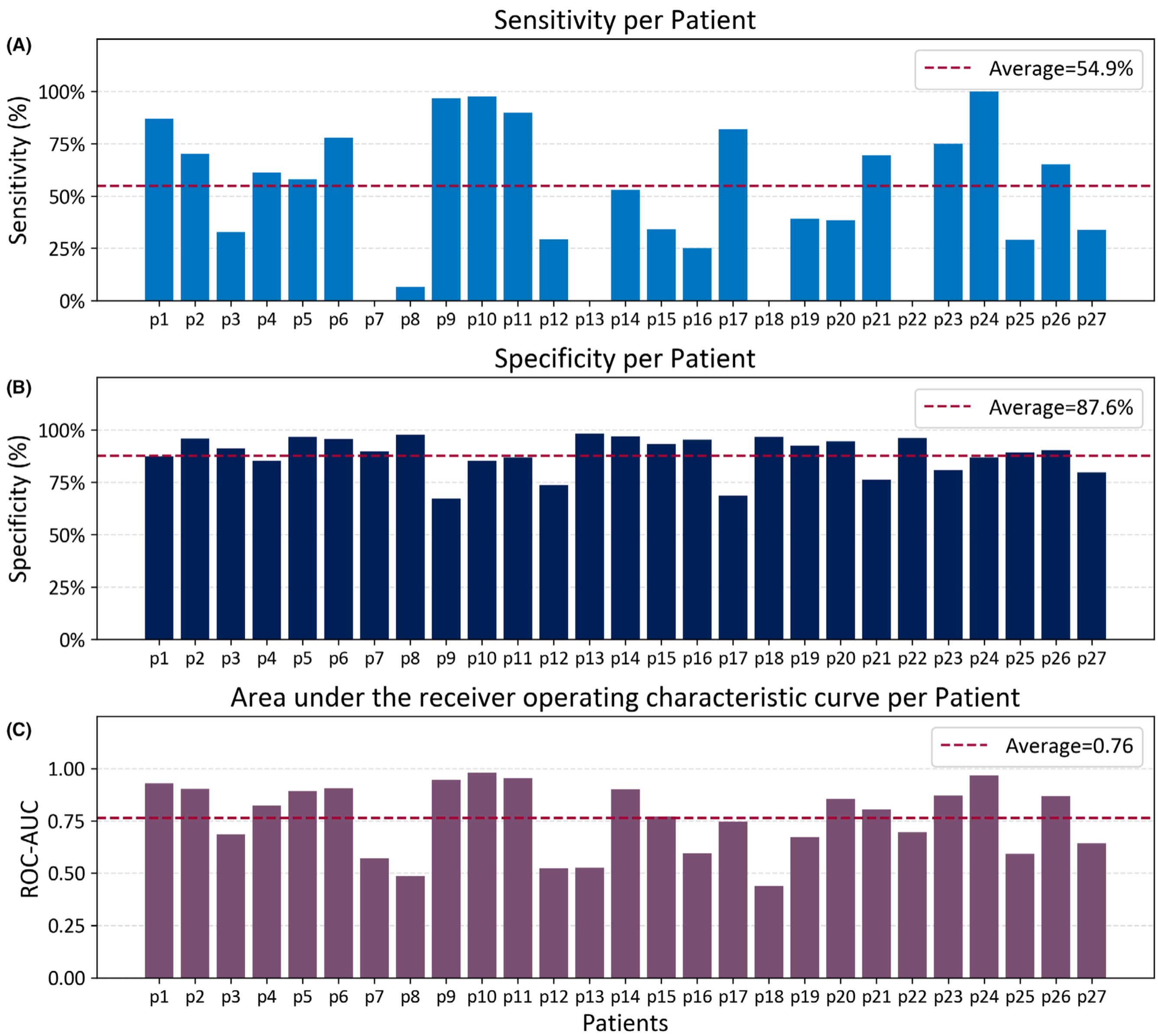

FIGURE 4 Epoch-based evaluation metrics per patient using a 10-second window with a 5-second step. ROC-AUC, area under the receiver operating characteristic curve.

distributed (Shapiro–Wilk test, $p<0.05$). In both cases, Mann–Whitney *U*-tests indicated no statistically significant difference between the groups (Threshold 0.65: $U=73.5000$, $p=0.5560$; Threshold 0.85: $U=87.0000$, $p=0.9380$).

In addition to focal seizures, 9 patients had FBTCS. Using the same trained XGBoost model with a firing power threshold of 0.65, all 17 seizures experienced by these patients were correctly detected. With the threshold of 0.85, only 2 seizures were missed, resulting in a sensitivity of 88%.

4 | DISCUSSION

In this study, we explored the feasibility of detecting FIAS using a biometric shirt that records respiratory, ECG, and accelerometry signals. We recruited patients admitted to the CHUM EMU and annotated seizures based on video-electroencephalography recordings. Features were extracted from raw signals using varying window sizes and steps, and an XGBoost classifier was trained and tested using a nested leave-one-subject-out cross-validation. Our methodology included a firing power regularization method to reduce false alarms. We recorded 113 FIAS from 27 patients, resulting in an average sensitivity of 66%, a TiW of 15%, and a FAR of 30 per 24 hours with a firing power threshold of 0.65. With a threshold of 0.85, we achieved a FAR of 21 per 24 hours with a lower sensitivity of 57%. In addition to FIAS, patients experienced FBTCS, FAS, and FUAS. While our algorithm was not trained for the detection of FBTCS, it detected most of them, with

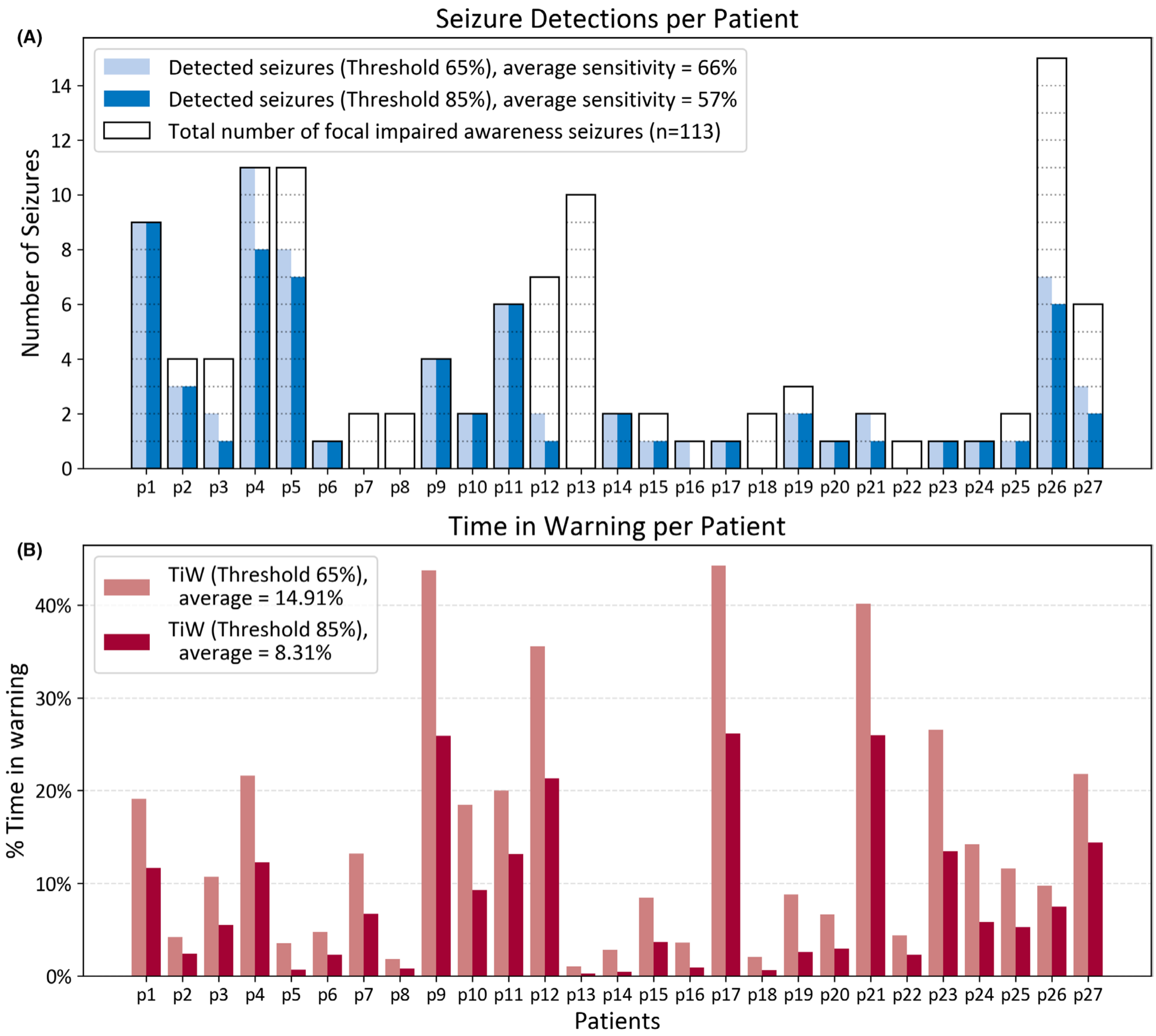

FIGURE 5 Event-based evaluation metrics per patient using a 10-second window with a 5-second step, and the firing power post-processing method with thresholds of 0.65 and 0.85. (A) Number of detected seizures compared to the total number of focal impaired awareness seizures per patient. (B) Percentage of time in false warning (TiW) per patient.

sensitivities of 100% (threshold of 0.65) and 88% (threshold of 0.85) calculated on the 17 FBTCS experienced by patients included in our analysis. Alarms generated during non-FIAS were considered false alarms, including FAS and FBTCS. FAR would not be significantly reduced if we considered those valid detections.

Traditional seizure detection methods often rely on EEG, which, while effective, is intrusive and impractical for continuous patient monitoring. Recent studies explored seizure detection using multimodal recordings acquired with wearable devices, such as the Embrace smart watch.[16,27,28] The biometric shirt used in this work provides a comfortable, non-stigmatizing, and user-friendly solution for long-term seizure monitoring. Its lightweight and discreet appearance makes it more acceptable for continuous use over weeks or months, particularly in ambulatory settings. To ensure a high-quality ECG signal, minor regular maintenance is required. Specifically, the shirt must be properly adjusted to the body, and a hydrating cream should be applied to the textile electrodes to maintain effective skin contact. Unlike wrist-worn devices, the shirt also captures respiratory signals, providing an additional modality to capture subtle physiological changes. This multimodal approach could help increase the robustness and accuracy of seizure detection for continuous monitoring.

We implemented a nested leave-one-subject-out cross-validation for performance evaluation. This technique ensures that our model is tested rigorously, simulating real-world scenarios where the model encounters unseen patients. By leaving one subject out for testing while training on the rest and nesting this process within another cross-validation loop for parameter tuning, we significantly reduced the risk of overfitting and ensured that our model generalizes well across different individuals. While this method reduces bias in model evaluation, the study was conducted in a single clinical center with a relatively small sample size (n = 27 patients). As such, future work should include validation on an independently acquired test dataset to formally assess our model's generalizability to other patient populations and different clinical settings.

To contextualize our results, we compared our model's performance with recent studies on wearable-based seizure detection. Yu et al.[29] trained a convolutional neural network long short-term memory (CNN-LSTM) based model using 900 seizures recorded from 166 patients with a wrist-worn wearable device measuring EDA, BVP, and acceleration. The dataset included various seizure types such as FBTCS, FAS, and FIAS, representing multiple semiologies. Combining acceleration and BVP, the proposed algorithm reached a ROC-AUC of 0.79, a sensitivity of 83.9%, and a false positive rate (FPR) of 35.3%.[29] While the authors did not report the corresponding FAR, our proposed algorithm performed better in terms of FPR (10%). Jeppesen et al.[30] explored seizure detection using ECG data extracted from a wearable ECG patch. They selected patients with a significant heart rate increase during seizures and achieved a sensitivity of 76% for focal seizures without bilateral generalization. However, the authors excluded 40%–50% of patients considered non-responders (less than 50 beats/min increase during seizures).[30,31] Patient selection allowed us to reach a FAR of 0.9/24 h. Böttcher et al.[18] used a wrist-worn device that measures BVP, EDA, and accelerometry. The authors selected focal seizures with motor components, and they achieved a 75% sensitivity with a FAR of 13.4/24 h. In our work, we did not perform patient pre-selection and reached a sensitivity of 66% over 27 patients and 113 seizures. By removing the 5 patients whose seizures were not detected at all, we achieved a sensitivity of 81% for the remaining 22 patients (96 seizures).

When interpreting our findings in comparison to previous studies, it is important to consider that we applied a 4-minute refractory period during evaluation; that is, alarms occurring during this time window were merged and counted as a single alarm. This choice was intended to reduce redundant alarms, potentially enhancing user experience in certain clinical or residential settings. However, longer refractory periods might artificially lower the counted number of false alarms, making direct comparisons with studies that use shorter, or no refractory periods less straightforward. To address this, we also report the percentage of time in false warning relative to the total recording duration, a metric that is less affected by the length of the refractory period.

In this study, we used the XGBoost classifier, a parallel tree boosting algorithm, due to its good performance on structured input and limited data, as well as its resistance to overfitting. This tree-based method handles heterogeneous hand-crafted features in a computationally efficient manner and has proven robust in various biomedical applications. Nevertheless, deep learning approaches, such as convolutional or recurrent neural networks, present promising alternatives for seizure detection from multimodal sensor data. These methods can capture and learn complex nonlinear patterns directly from raw input signals. With larger and diverse datasets, deep neural networks could potentially enhance seizure detection and support the development of more individualized seizure monitoring solutions.

Detecting focal seizures is particularly challenging due to the varied semiology of this type of seizure. In this work, we did not select patients based on the specific characteristics of their clinical or physiological manifestations, such as heart rate increase or motor components, which adds to the complexity of detection. Despite these challenges, our methodology demonstrated promising performance in certain cases (sensitivity of 100% for 13 patients). In contrast, some patients would not be candidates for this device, and removing the 5 patients for whom no seizures were detected increased the average sensitivity to 81%. Given known anatomical and physiological differences between sexes, we examined whether sensitivities were different between male and female patients. Although we observed a slight variability between both patient groups, these differences were not statistically significant ($p > 0.05$). Other clinical factors, such as seizure semiology or the location of the epileptic focus, could also influence detection rates. However, due to our limited sample size, we were not able to perform a formal analysis to assess their impact. Future studies with larger and more diverse patient cohorts are required to explore these potential sources of variability in a systematic manner.

Although not within the scope of this work, future efforts may explore broader applications of the biometric shirt beyond seizure detection. In fact, psychogenic non-epileptic seizures (PNES) often mimic epileptic events in their outward clinical manifestations. Differentiating PNES from epileptic seizures remains a highly relevant clinical challenge. While this study exclusively focused on confirmed epileptic seizures, the analysis of multimodal electrophysiological signals may offer valuable information to distinguish PNES from epileptic seizures. Previous studies have shown distinct autonomic profiles between PNES and epileptic seizures, suggesting that biometric wearable devices could support differential diagnosis in ambiguous cases.[32,33] Future work could involve cohorts with both epilepsy and PNES groups to explore this potential application.

Variability in sensitivity between patients highlights the potential for personalized seizure detection algorithms tailored to individual physiological and clinical characteristics. Given the heterogeneity of focal seizures, including differences in seizure semiology, epileptic brain networks, and electrophysiological signatures, a generalized one-fits-all solution may be suboptimal. In this context, personalized models may offer a more effective strategy to capture patient-specific patterns, improve sensitivity, and reduce false alarms. For instance, Böttcher et al.[18] demonstrated that personalized models trained on data from individual patients outperformed generalized models in detecting focal onset motor seizures using wearable sensors. These findings support the rationale that patient-tailored algorithms could enhance the accuracy, reliability, and usability of seizure monitoring, particularly in real-world applications.

Our study has limitations. The false alarm rate remains very high, which could lead to alarm fatigue and reduced trust in the device. There was considerable variability in algorithm performance across patients, indicating that a one-size-fits-all approach may not be effective for the detection of FIAS using wearable devices. The study was conducted in a controlled hospital environment, and real-world conditions may present additional challenges.

Future research aims to improve performance by exploring patient-specific approaches to better tailor the detection algorithms to individual physiological profiles. The algorithm should be implemented in real time and tested prospectively.

5 | CONCLUSION

While our study shows promise for the detection of FIAS using a biometric shirt, significant challenges remain. Addressing patient variability, reducing false alarms, and ensuring practicality for everyday use are critical areas for future research. Furthermore, multicentric studies and real-time prospective testing will be essential to validate and enhance the applicability of these technologies for a broader patient population.

AUTHOR CONTRIBUTIONS

Each author contributed to this article. Jérôme St-Jean: Conceptualization (equal); formal analysis (lead); methodology (lead); software (equal); visualization (equal); investigation (equal); and writing – original draft (lead). Oumayma Gharbi: Software (equal); visualization (equal); investigation (equal); and writing – review and editing (lead). Dènahin Hinnoutondji Toffa: Data curation (equal). Thi Phuoc Yen Tran: Data curation (equal). Manon Robert: Investigation (equal) and resources (lead). Dang Khoa Nguyen: Conceptualization (equal); data curation (equal); supervision (equal); funding acquisition (equal); and writing – review and editing (equal). Elie Bou Assi: Conceptualization (equal); methodology (supporting); supervision (equal); validation; funding acquisition (equal); and writing – review and editing (equal).

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CONFLICT OF INTEREST STATEMENT

None of the authors has any conflict of interest to disclose. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

DATA AVAILABILITY STATEMENT

The datasets presented in this article are not readily available because sharing of raw data is conditional to approval by our Institution's Research Ethics Board. Requests to access the datasets should be directed to EBA, elie.bou.assi.chum@ssss.gouv.qc.ca.

ETHICS STATEMENT

The study received approval from the Institutional Ethics Research Board (18.091). Written informed consent was obtained from each participant prior to their participation.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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